Exercise Sheet 5

Community Detection

5942UE Network Science

Prof. Dr. Michael Granitzer Dr. Kanishka Ghosh Dastidar Dr. Jelena Mitrovic

5.A Basic Measures and Modularity

Exercise 5.A.1: Community Density by Hand

Graph: A, B, C, D, E, F, edges A–B, A-C, B–C, B–D, D–E, D–F, E–F.

Two candidate partitions:

- **P1:** A, B, C, D and E, F
- **P2:** A, B, C and D, E, F

For each partition:

1. Compute **internal edge density**: $|\text{edges inside}|$ / $|\text{possible edges inside}|$.
2. Compute **external edge density**: $|\text{crossing edges}|$ / $|\text{possible crossing edges}|$.
3. Which partition better separates the network? Justify with numbers.
4. What is the "ground truth" community structure visible from inspection?

Exercise 5.A.2: Modularity — Concept and Computation

Modularity Q measures whether communities are denser than expected by chance:

$$Q = \sum_c \left(\frac{l_c}{m} - \left(\frac{d_c}{2m} \right)^2 \right)$$

where l_c = edges inside community c, d_c = sum of degrees in community c, m = total edges.

1. For **P2** from Exercise 5.A.1, compute Q by hand (use $m = 7$).
2. What does $Q = 0$ mean? What does $Q = 1$ mean? What range is typical for real networks?
3. If you randomly shuffled community assignments, what would you expect Q to approach?

Exercise 5.A.3: Community Detection in Python

Using `nx.karate_club_graph()`:

1. Apply greedy modularity maximisation: `nx.community.greedy_modularity_communities(G)`.
2. Compute the modularity score of the result using `nx.community.modularity(G, communities)`.
3. Compare to the known faction labels (`G.nodes[n]["club"]`). What fraction of nodes are assigned to the "correct" faction?
4. Visualise: plot the network with nodes coloured by detected community.

5.B Girvan-Newman Algorithm

Exercise 5.B.1: Girvan-Newman Steps

Graph: 6 nodes, two triangles $1, 2, 3$ and $4, 5, 6$ connected by single edge $3–4$.

1. Compute **edge betweenness** for all edges. Which edge has the highest value?
2. Remove that edge. What happens to the graph?
3. Which edges now have the highest betweenness after the removal?
4. Why is edge $3–4$ guaranteed to have the highest betweenness in this structure?

Exercise 5.B.2: Girvan-Newman in Practice

Using `nx.karate_club_graph()`:

1. Apply the Girvan-Newman algorithm: `nx.community.girvan_newman(G)`.
2. Extract the partition at the 2-community level. Does it recover the known factions?
3. Compare faction recovery accuracy to the greedy modularity result from Exercise 5.A.3.
4. Plot the edge betweenness of the top-5 edges at the initial state and after 1 removal step.

5.C Hierarchical Clustering

Exercise 5.C.1: Dendrogram Interpretation

Consider the 6-node two-triangle graph from the Girvan-Newman exercise.

1. What does the **height** of a merge step in a dendrogram represent?
2. In the dendrogram for this graph, at what level do the two triangles merge into a single cluster?
3. How would you choose the "right" number of clusters from a dendrogram?
4. What is the difference between using **direct-link similarity** vs. **neighbourhood-overlap similarity** as the merge criterion?

Exercise 5.C.2: Hierarchical Clustering in Python

Using `nx.karate_club_graph()` and `scipy`:

1. Compute pairwise **shortest-path distances** between all nodes. Use this as the distance matrix.
2. Apply hierarchical clustering with average linkage: `scipy.cluster.hierarchy.linkage(...)`.
3. Plot the resulting dendrogram.
4. Cut the dendrogram to obtain 2 clusters. What fraction of nodes match the known factions?